

A Geospatial Ex-Ante Profitability Model for Basalt Enhanced Rock Weathering in Sub-Saharan Africa

Working Paper

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2026-05-15

Abstract

Enhanced rock weathering (ERW) on agricultural soils is among the most-scalable durable carbon-dioxide-removal pathways currently available, yet its economics are dominated by spatial variation in feedstock chemistry, logistics, agronomic response, and policy parameters. Region-specific ex-ante models are scarce. We present a geospatial model that estimates per-pixel gross margin of basalt-ERW deployment across 47 Sub-Saharan African countries and 23 SPAM v2 crops at approximately one-square-kilometre resolution. The model decomposes profitability into three causal streams — biogeochemical CDR, agronomic yield uplift, and economic cost — that converge on a single gross-margin equation modulated by six decision variables (feedstock chemistry, grain size, allocation rule, carbon price, monitoring-reporting-and-verification cost, and discount rate). It composes nine methodological blocks: a working grid from SoilGrids-Africa and SPAM v2; basalt application rates derived from lime-equivalent demand; a climate-and-pH-scaled CDR surface with an aridity-driven pedogenic-carbonate deduction; a delivered-basalt-price surface from GLiM v1 source rocks and the Malaria Atlas Project motorised friction surface; country-resolved electricity tariffs and grid carbon intensity; a Bayesian-hierarchical yield-uplift posterior anchored on a curated ERW-trial dataset; and a supply-constrained deployment ranking. We illustrate the model with first-cut indicator maps — soil pH, basalt application rates, CDR efficiency, delivered feedstock price — and discuss the structural gaps that currently constrain a fully closed-loop run, principally the omission of industrial alkaline by-products as feedstock, the prior-dominated trial dataset, and the use of mean annual precipitation rather than a full aridity index for the pedogenic-carbonate deduction. The model is implemented in R using the terra, geodata, and brms packages. Code and companion documentation are released alongside this paper.

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1 1. Introduction

The 2023 IPCC Sixth Assessment Report’s Working Group III synthesis [1] establishes that limiting warming to 1.5–2 °C requires not only deep emission cuts but also gigaton-scale **carbon dioxide removal (CDR)** by mid-century. Among the candidate CDR pathways with non-trivial mitigation potential and reasonable technological readiness, enhanced rock weathering on agricultural soils stands out: it is mechanism-known, scalable in principle to multiple gigatons per year, and offers the rare property of plausibly *paying for itself* through agronomic co-benefits in regions with acidic, weathered soils [2, 3].

The basic mechanism is straightforward. Finely ground silicate rock — in practice almost always basalt or a closely related mafic rock — is applied to cropland. Carbonic acid in soil water dissolves

the silicate, releasing alkalinity in the form of bicarbonate ions. Over time, the alkalinity is exported to rivers, then to the ocean, where it remains stable on geological timescales [4]. The agronomic side effect is a near-lime-equivalent reduction in soil acidity, which is itself a binding constraint on cropland productivity across much of the humid tropics.

The headline numbers for ERW are encouraging. Beerling et al. [2] estimate that deploying basalt across global croplands could remove 0.5–2 Gt CO₂ per year by 2050, with the highest per-hectare potential in the warm-humid tropics. Tropical cropland — of which Sub-Saharan Africa contains roughly 220 Mha — sits at the top of the priority list under any climate-driven ranking. Field trials in the US Corn Belt have measured a cumulative 10.5 ± 3.8 t CO₂ per hectare over four annual applications (50 t·ha⁻¹·yr⁻¹) with a 12–16 % maize-yield uplift [5]. A 2024 smallholder trial in Kisumu County, Kenya, reported a 71 % first-year and 79 % second-year maize-yield increase from a single 20 t/ha basalt application on a soil at pH 6.4, yielding a net farmer benefit of approximately \$326 per hectare [6]. By contrast, a 2025 Swiss-vineyard trial measured only about 100 kg CO₂ per hectare per year [7], an order of magnitude lower than the higher published rates. The Swiss result does not invalidate ERW; rather, it confirms that climate, soil pH, and drainage are the dominant controls on realised rates. Warm, mildly acidic, well-drained, vegetated systems sit at the high end of the response distribution — exactly the agroecological space most of cropland Sub-Saharan Africa occupies.

What is missing from the literature, however, is a **region-specific ex-ante profitability model** at the resolution at which deployment decisions are actually made. Existing global or continental models [2, 3, 8] are either too coarse or too aggregated to inform location-specific deployment, MRV protocols, or policy. Existing economic analyses [9, 10] typically assume a uniform delivered-feedstock price and uniform grinding cost, both of which vary by a factor of 5 or more across Sub-Saharan Africa. This paper presents a first attempt to fill this gap. We build a per-pixel, decision-variable-aware profitability model for basalt-ERW deployment across Sub-Saharan Africa, and present its conceptual framework, methods, and illustrative outputs.

The contribution of the paper is threefold. First, we propose a three-stream decomposition of the gross-margin equation that cleanly separates the biogeochemical, agronomic, and economic mechanisms without losing the structural cross-link between them (the agronomic lime-equivalent demand sets the basalt application rate that drives the biogeochemical stream). Second, we operationalise this decomposition at approximately one-kilometre resolution across forty-seven Sub-Saharan African countries, with feedstock chemistry, grain size, allocation rule, carbon price, MRV cost, and discount rate all exposed as user-settable decision variables. Third, we make the model and its source code public, with continuously updated documentation, so that future trial evidence and policy parameters can be incorporated as they become available.

The remainder of the paper is organised as follows. Section 2 presents the conceptual framework. Section 3 catalogues the data sources. Section 4 specifies the methodological blocks and their governing equations. Section 5 documents the pipeline architecture and implementation. Section 6 presents illustrative first-cut outputs and their interpretation. Section 7 discusses limitations and outlines a roadmap. Section 8 concludes.

2. Conceptual framework

We model the **per-pixel gross margin** of basalt-ERW deployment as a sum of two revenue streams and one cost stream:

$$\pi(x, y \mid \text{regime, allocation}) = R_{\text{CDR}}(x, y) + R_{\text{agro}}(x, y) - C(x, y). \quad (1)$$

Here (x, y) indexes a working-grid pixel and the conditioning arguments index the temporal regime (year-1, net-present-value at 10 %, equilibrium) and the allocation rule (LiTAS-targeted versus uniform 10, 20, or 50 t ha⁻¹). The three terms each correspond to a causal stream with its own physical or economic mechanism. The streams converge on equation (1) but interact only through one well-defined cross-link, which we describe below. The framework is summarised graphically in Figure 1.

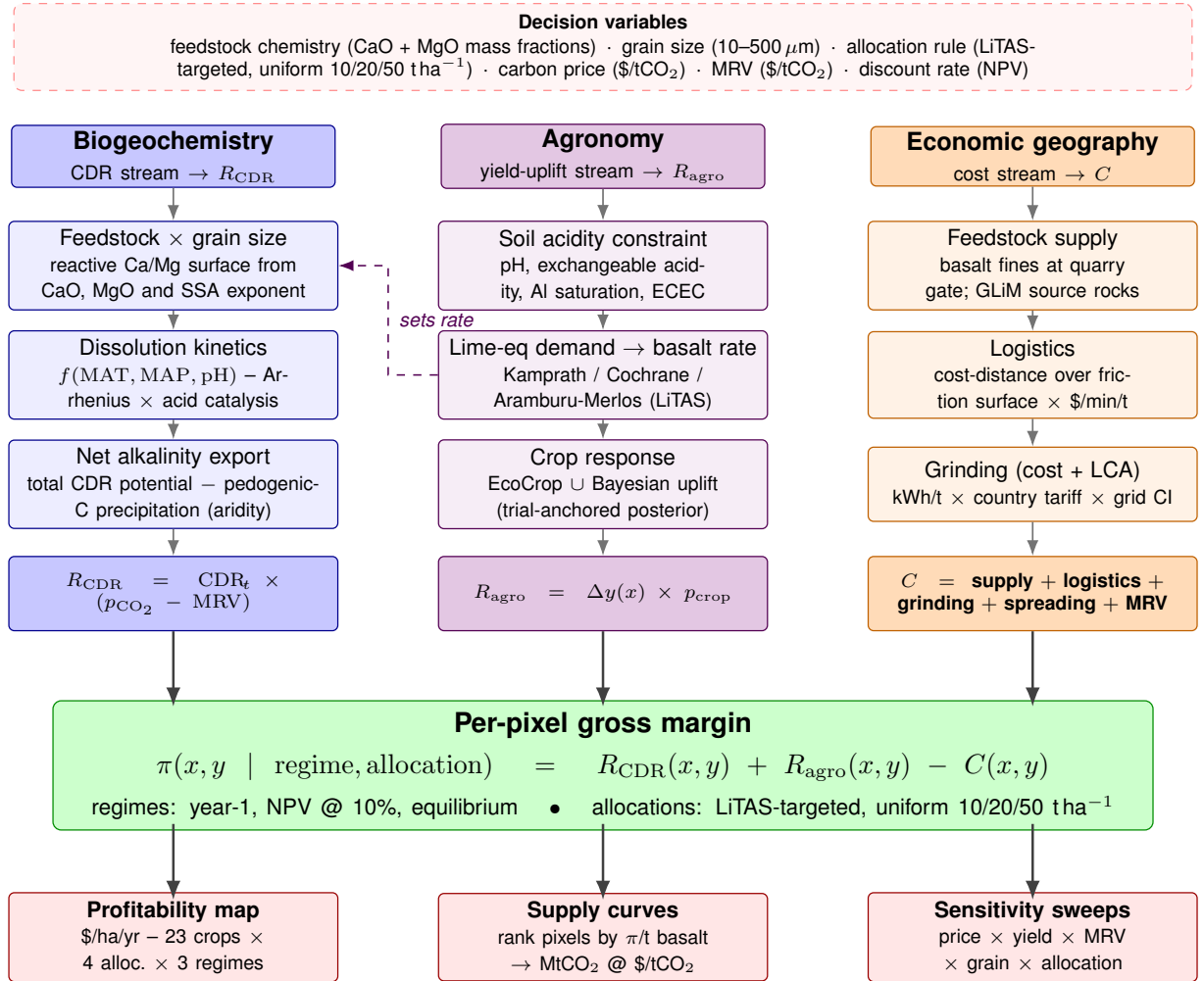


Figure 1: Conceptual framework: three causal streams (biogeochemistry, agronomy, economic geography) converging on the per-pixel gross-margin equation, modulated by six decision variables and producing three downstream uses. The dashed cross-link captures the structural insight that agronomic lime-equivalent demand sets the basalt application rate that the biogeochemistry stream then dissolves.

2.1 2.1 Biogeochemical stream

The biogeochemical stream computes the carbon-dioxide-removal revenue, R_{CDR} . It begins with the **reactive cation surface** of the feedstock, which is a function of the feedstock chemistry (CaO and MgO mass fractions) and grain size:

$$\text{SSA}(d) \propto d^{-\beta}, \quad \beta = 0.5. \quad (2)$$

We follow Strefler et al. [3] in using $\beta = 0.5$ for the surface-area exponent and $\alpha = 1.2$ for the grinding-energy exponent. The reactive surface is then dissolved at a rate that combines an **Arrhenius-style temperature term**, a **moisture term**, and a **triangular pH-acid-catalysis term** peaked near pH 5:

$$k_{\text{diss}}(T, P, \text{pH}) = k_{\text{ref}} \cdot f_T(T) \cdot f_P(P) \cdot f_{\text{pH}}(\text{pH}), \quad (3)$$

with reference values $T_{\text{ref}} = 11$ °C (United-States Corn-Belt anchor) and $P_{\text{ref}} = 1000$ mm. The reference reactive fraction $k_{\text{ref}} = 0.20$ is calibrated from the Lewis et al. [11] basalt-weathering compilation. Realised CDR is reduced by an **aridity-driven pedogenic-carbonate fraction** ϕ_{ped} , which removes the alkalinity that precipitates as secondary carbonate locally rather than exporting as oceanic bicarbonate:

$$\text{CDR}_t(x, y) = \text{CDR}_{\text{stoich}}(x, y) \cdot k_{\text{diss}} \cdot (1 - \phi_{\text{ped}}(\text{MAP})). \quad (4)$$

We bin ϕ_{ped} using the UNEP-1992 aridity bands keyed on mean annual precipitation [12]; this is a first-version proxy for the full aridity index $\text{AI} = P/\text{PET}$. CDR revenue is then

$$R_{\text{CDR}}(x, y) = \text{CDR}_t(x, y) \cdot (p_{\text{CO}_2} - c_{\text{MRV}}), \quad (5)$$

with $p_{\text{CO}_2} = \$150/\text{tCO}_2$ as a baseline carbon price (the 2024–25 mid-market for verified ERW credits) and $c_{\text{MRV}} = \$30/\text{tCO}_2$ as a baseline MRV cost (sampling, laboratory, verification, and registry fees).

2.2 2.2 Agronomic stream

The agronomic stream computes the yield-uplift revenue, R_{agro} . It begins with the **soil-acidity constraint**: pH, exchangeable acidity, aluminium saturation, and effective cation-exchange capacity, all from SoilGrids-Africa [13]. The constraint is converted to a **lime-equivalent demand** via one of three recipes available in the `limer` R package [14]: Kamprath, Cochrane, or Aramburu-Merlos LiTAS [15]. The lime demand is then converted to a basalt application rate using the feedstock chemistry:

$$\text{rate}_{\text{basalt}}(x, y) = \frac{\text{lime}_{\text{CaCO}_3}(x, y)}{\eta \cdot (w_{\text{CaO}} + w_{\text{MgO}} \cdot M_{\text{CaO}}/M_{\text{MgO}})}, \quad (6)$$

where w_{CaO} and w_{MgO} are the feedstock-chemistry mass fractions (default 10 % and 7 %) and η is the Lewis-2021 reactive fraction. **This rate is the cross-link with the biogeochemical stream:** it is the same quantity that drives equations (4) and (5).

Crop response per pixel is the maximum of two surfaces:

- An **EcoCrop suitability** $S_{\text{EC}}(\text{crop}, \text{pH}, \text{Al}_{\text{sat}}) \in [0.2, 1]$, computed via Recocrop [16] from the pH and acidity-saturation tolerance tables of the 23 SPAM v2 crops [17].
- A **Bayesian-hierarchical yield-uplift posterior** $\Delta y_{\text{B}}(\text{crop}, \text{pH}, \text{MAT}, \text{MAP}, d) \mid \mathbf{x}$ from a brms model [18] fitted on a curated ERW-trial dataset (currently 14 published trials).

When the Bayesian surface is available, the model prefers it; otherwise it falls back to the EcoCrop surface. Yield revenue is then

$$R_{\text{agro}}(x, y) = \Delta y(x, y) \cdot p_{\text{crop}}(c), \quad (7)$$

with $p_{\text{crop}}(c)$ the FAOSTAT 2016–2020 producer-price average for crop c in the country containing pixel (x, y) .

2.3 2.3 Economic-geography stream

The economic-geography stream computes the total cost, C . It composes four components:

$$C(x, y) = c_{\text{quarry}} + c_{\text{logistics}}(x, y) + c_{\text{grinding}}(x, y) + c_{\text{operations}}. \quad (8)$$

- c_{quarry} is the **quarry-gate price** of basalt fines, fixed at \$10 per tonne (the by-product price of fines too small for aggregate use).
- $c_{\text{logistics}}(x, y) = \tau(x, y) \cdot \rho$, where $\tau(x, y)$ is the **minute-of-travel cost-distance** from pixel (x, y) to the nearest basalt source, accumulated over the Malaria Atlas Project 2019 motorised friction surface [19]; and $\rho = \$0.04/\text{min}/\text{t}$ is the unit haulage cost (30-tonne truck at \$80 per hour). We cap the haul at $L_{\text{max}} = 1000$ km, beyond which truck-diesel lifecycle emissions would erase the credit.
- $c_{\text{grinding}}(x, y) = E_{\text{grind}} \cdot p_{\text{elec}}(x, y)$, where E_{grind} is the grinding energy intensity (kWh per tonne, proportional to grain size raised to the Streffer exponent $\alpha = 1.2$) and $p_{\text{elec}}(x, y)$ is the country industrial-electricity tariff.
- $c_{\text{operations}} = c_{\text{spreading}} + c_{\text{MRV}}$ with $c_{\text{spreading}} = \$8/\text{t}$ on-farm and c_{MRV} already deducted from the carbon-revenue side in equation (5).

A separate **lifecycle carbon term**, $\text{LCA}(x, y) = \text{gridCl}(x, y) \cdot E_{\text{grind}}$, is deducted from gross CDR before equation (5) is applied; this is the spatial-variation channel through which grid carbon intensity matters.

2.4 2.4 Decision variables

Six dials are exposed at the framework boundary and propagate inwards:

1. **Feedstock chemistry** — CaO and MgO mass fractions (w_{CaO} , w_{MgO}).
2. **Grain size** — d , sweepable from 10 to 500 μm .

3. **Allocation rule** — LiTAS-targeted versus uniform 10, 20, or 50 t ha⁻¹.
4. **Carbon price** — p_{CO_2} , default \$150 per tCO₂.
5. **MRV cost** — c_{MRV} , default \$30 per tCO₂.
6. **Discount rate** — r , default 10 % per year (NPV regime only).

Five sensitivity sweeps are implemented as standard outputs: a crop-price by basalt-price by carbon-price grid; a yield-uplift sweep from 1.0× to 2.5×; an MRV-cost sweep from 0 to 80 \$/tCO₂; a grain-size sweep over 10, 30, 50, 100, 200, and 500 µm; and an allocation-rule sweep across the four allocations. All sweeps are run across the three temporal regimes and all 23 crops.

3. Data

The model consumes ten external datasets, listed in Table 1. All sources are public or available through standard R interfaces, with one exception (the FAOSTAT producer-price tables, which require a manual CSV download).

Table 1: Data sources used by the model. {#tbl:data}

Dataset	Version	Access	Used in
GADM	v4.1	<code>geodata::world()</code>	B1 working grid
SoilGrids-Africa	iSDA, 30 arc-sec	<code>geodata::soil_af()</code>	B1 working grid
QED cropland mask	latest	<code>geodata::cropland(source="1000k")</code>	B1 working grid
SPAM	v2 (Africa)	<code>geodata::crop_spam()</code>	B1 working grid
EcoCrop parameters	curated	<code>Recocrop::ecocropPars()</code>	B6 EcoCrop arm
WorldClim	v2.1 (MAT, MAP)	<code>geodata::worldclim_global()</code>	B4 CDR, B6 Bayesian
GLiM lithology	v1.0 / 1.1 (GDB)	PANGAEA download [20]	B4 logistics
MAP friction surface	motorised, nominal 2019	<code>malariaAtlas(dataset_id = Ex-plorer__2020_motorized_friction_surface)</code>	B4 logistics
FAOSTAT producer prices	2016–2020 average	manual CSV	B7 profitability
Country energy (tariff + grid CI)	2022–2024	hard-coded CSV in <code>erw-energy-country.R</code>	B5 energy
ERW trial dataset	curated by authors	<code>erw/erw-trial-data.csv</code> (14 rows)	B6 Bayesian arm

The working spatial grid is the SoilGrids-Africa 30-arc-second grid aggregated ten-fold by spatial mean, yielding a resolution of approximately $0.083^\circ \approx 9$ km (746 × 828 cells across 47 Sub-Saharan African countries). This coarsening is necessary both for computational tractability and to match the effective decision unit of farm-level deployment, which is rarely finer than 1 km².

Notably, our feedstock-source inventory — GLiM v1 basic-volcanic and basic-plutonic polygons — represents only a subset of the literature’s full candidate-feedstock set [3, 22]. We discuss this omission, and the others, in Section 7.

4. Methods

This section specifies the nine methodological blocks introduced in Section 2. Each block corresponds to one or more scripts in the implementation (Section 5); we note the script(s) but defer

detailed code-level documentation to the companion model documentation.

4.1 4.1 Working grid (B1)

The working grid is built from the union of GADM v4 polygons for 47 Sub-Saharan African countries, masked to QED cropland, and intersected with the SoilGrids-Africa property stack (pH, bulk density, exchangeable acidity, exchangeable K, Ca, Mg, Na bases). Effective cation-exchange capacity (ECEC) and acidity saturation are derived inline. The SPAM v2 harvested-area, production, and yield rasters for 23 crops are resampled to the same grid. All downstream rasters are co-registered on this grid.

4.2 4.2 Basalt application rate (B2)

Lime-equivalent demand in t CaCO_3 per hectare is computed via the `limer::limeRate` function, which implements four recipes: Kamprath (correction of exchangeable aluminium), Cochrane (correction of aluminium saturation), Aramburu-Merlos year-1 (LiTAS), and Aramburu-Merlos maintenance. The chosen recipe (default: LiTAS) is converted to basalt rate via equation (6). Uniform 10, 20, and 50 t ha^{-1} counterfactual rates are computed in parallel for the allocation-rule sensitivity sweep. The output is four LiTAS-style basalt-rate rasters plus three uniform rasters per allocation.

4.3 4.3 CDR surface (B3)

For each pixel, the stoichiometric CDR potential per tonne basalt is computed from w_{CaO} and w_{MgO} and the $\text{Ca}^{2+}/\text{Mg}^{2+}$ molar masses, assuming complete dissolution and conservative ocean export. Per-tonne CDR is then scaled by equation (3) to produce a per-tonne effective-CDR surface, multiplied by the basalt rate from B2 to produce a per-hectare CDR yield, and discounted by the ϕ_{ped} fraction from equation (4). The output is the per-tonne CDR surface, the per-hectare CDR-yield surface, the climate factor surface, and the pedogenic-carbonate fraction surface, each at each of the four LiTAS allocations and the three uniform allocations.

4.4 4.4 Logistics surface (B4)

Basalt source rocks are extracted from the GLiM v1 lithology database as polygons whose `xx` attribute belongs to the set `{vb, py, pb}` (basic volcanic, pyroclastic, basic plutonic). Globally, these classes cover 107,597 polygons; after cropping to the Sub-Saharan-African extent, 6,455 polygons remain, contributing 1,399,005 mafic-source cells (approximately 1.9 % of the SSA cropland-context grid). We compute a per-pixel travel-time-to-source surface, $\tau(x, y)$, using `terra::costDist` accumulating over the MAP 2019 motorised friction surface [19]. To enable `terra`'s signature, source pixels are marked with a sentinel value of `-1` in a copy of the friction surface and passed as `target = -1`. Travel time is capped at 1000 minutes (approximately 1000 km at the 60 km/h free-flow proxy). The delivered basalt price is then

$$p_{\text{delivered}}(x, y) = c_{\text{quarry}} + \tau(x, y) \cdot \rho, \quad (9)$$

which produces our final B4 surface. The delivered-price distribution across Sub-Saharan Africa has mean \$40.8 per tonne, range \$10–50 per tonne, with the upper cap binding for roughly half

of pixels — principally in West Africa and the Sahel, where neither basalt nor gabbro outcrops are common.

4.5 4.5 Country energy surface (B5)

Per-country industrial-electricity tariffs and grid carbon intensities are hard-coded in a CSV (sourced from IEA, Ember, and GET.invest 2022–2024) and rasterised onto the working grid by GADM country code. Pixels in countries without a tabulated value fall back to \$0.10 per kWh and 0.60 kg CO₂ per kWh. The output is two rasters: $p_{\text{elec}}(x, y)$ and $\text{gridCI}(x, y)$.

4.6 4.6 Crop response (B6)

The EcoCrop arm of B6 builds two suitability rasters per crop, one for pH and one for aluminium saturation, using the tolerance tables in `Recocrop : ecocropPars ( )`. Per-pixel suitability is clamped to the range $[0.2, 1]$ to avoid extreme zero values that would over-penalise marginal land.

The Bayesian arm fits a hierarchical regression of the form

$$\log(y_{\text{with}}/y_{\text{control}}) = \beta_0 + \beta_1 \cdot \text{rate} + \beta_2 \cdot \text{pH} + \beta_3 \cdot \text{MAT} + \beta_4 \cdot \text{MAP} + \beta_5 \cdot d + u_{\text{croptype}} + \varepsilon, \quad (10)$$

where u_{croptype} is a random intercept by crop-type group (cereal, legume, root-tuber-banana, commodity). With only 14 trial rows currently available, the posterior is prior-dominated; we treat this as a deliberately conservative framework that grows in evidence weight as new field-trial data is appended. Per-pixel posterior medians are applied to the working-grid covariates to produce yield-uplift surfaces $\Delta y_B(x, y)$ for each of the 23 crops.

4.7 4.7 Profitability (B7)

For each combination of allocation rule, temporal regime, and crop, the per-pixel gross margin is computed via equation (1). The temporal regime determines the CDR-realisation phasing (30 / 25 / 20 / 15 / 10 % over years 1 through 5; equilibrium runs assume complete dissolution) and, for the NPV regime, a 10-per-cent annual discount. The output is a profitability raster stack per (allocation, regime, crop), yielding $4 \times 3 \times 23 = 276$ rasters per run.

4.8 4.8 Supply-constrained deployment (B8)

For each pixel, the **shadow value** of basalt supply — the gross margin per tonne of basalt applied — is computed as the LP-dual on the basalt-supply constraint:

$$\text{shadow}(x, y) = \frac{\pi(x, y)}{\text{rate}_{\text{basalt}}(x, y)}. \quad (11)$$

Pixels are ranked descending by shadow value, and the cumulative deployed hectares and total CDR are computed as a function of an annual basalt-supply cap. The cap is swept over 1, 5, 10, 25, 50, 100, 250, 500 Mt per year, plus an unconstrained reference. The output is one supply curve per temporal regime, plus a deployment-mask raster at selected caps.

4.9 Sensitivity sweeps (B9)

Five sweeps are implemented; their axes are listed at the end of Section 2.4. Each sweep is a Cartesian product over the listed values; the output is a long-format CSV of total profitable hectares and gross margin under each combination of regime and crop.

5. Implementation

The model is implemented in R 4.5 using the `terra`, `geodata`, `sf`, `here`, `brms`, and `malariaAtlas` packages. The pipeline architecture is shown in Figure 2.

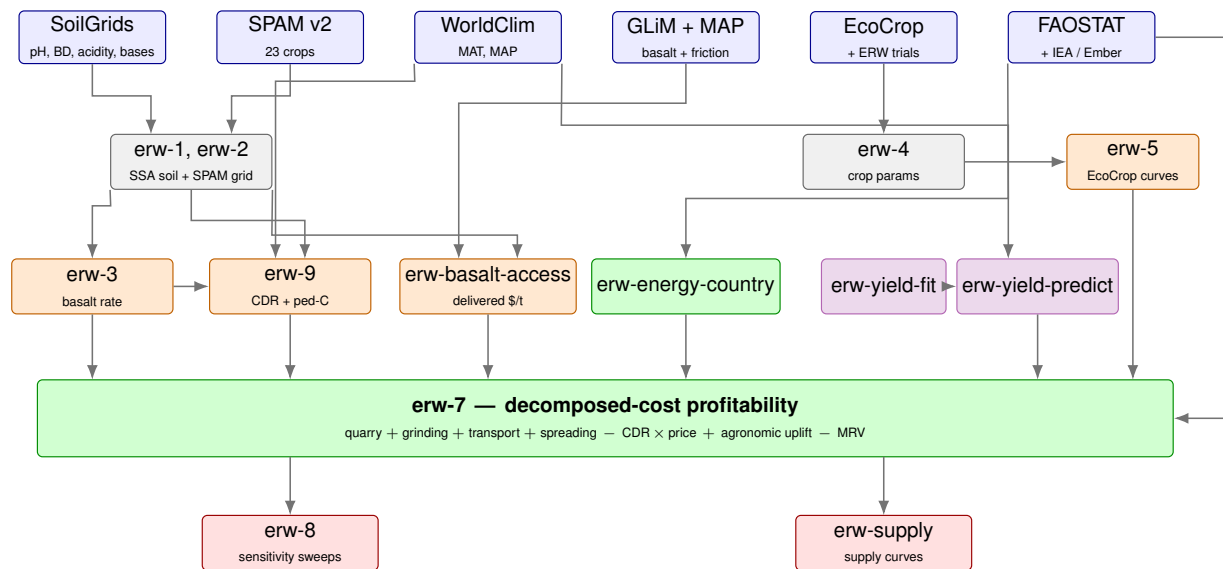


Figure 2: Pipeline architecture. Boxes are scripts (`erw-*.R`); arrows are data dependencies. Optional arms (`erw-basalt-access`, `erw-energy-country`, `erw-yield-predict`) feed `erw-7` when present; otherwise the pipeline falls back to flat constants and EcoCrop curves. All arrows are orthogonal and each horizontal segment lives on a dedicated y-track to avoid co-linear overlap.

The pipeline is layered: a data-input layer (six datasets) feeds a working-grid layer (`erw-1`, `erw-2`, `erw-4`, `erw-5`), which feeds a surface layer (`erw-3`, `erw-9`, `erw-basalt-access`, `erw-energy-country`, `erw-yield-fit`, `erw-yield-predict`). The surface layer converges on the profitability engine (`erw-7`), which feeds the downstream sensitivity sweeps (`erw-8`) and the supply-constrained ranking (`erw-supply`). Each script writes its outputs as GeoTIFFs to the working directory; downstream scripts read these GeoTIFFs as inputs, so individual stages can be re-run without re-running upstream stages.

Run-times on Apple Silicon (single-core) are dominated by the initial data-download stages (`erw-1` and `erw-2`, approximately 30 to 60 minutes when uncached) and by `terra::costDist` in `erw-basalt-access` (approximately 10 minutes). Other stages run in seconds to a few minutes.

Source code is at `<repo>/erw/`; companion documentation is at `<repo>/docs/erw-model.md` (or the compiled PDF, `docs/erw-model.pdf`); and an interactive browser-rendered view is at `<repo>/flows.html`.

6 Illustrative outputs

We present four selected indicator maps that together convey the model's first-order outputs and the spatial heterogeneity of the inputs. Twelve indicator maps are rendered in total by the companion script `erw/erw-plot-maps.R`; the remainder are reproduced in the model documentation.

6.1 Soil pH (input to B1)

Figure 3 shows the depth-weighted topsoil pH from SoilGrids-Africa, masked to cropland. Two zones dominate: a humid-tropical band from West Africa through the Congo basin to East Africa with pH typically below 6, and a semi-arid band across the Sahel and parts of southern Africa with pH typically above 7. The humid-tropical band is the region where lime-equivalent demand — and therefore basalt application rate, equation (6) — is highest, and where ERW makes both an agronomic and a CDR contribution.

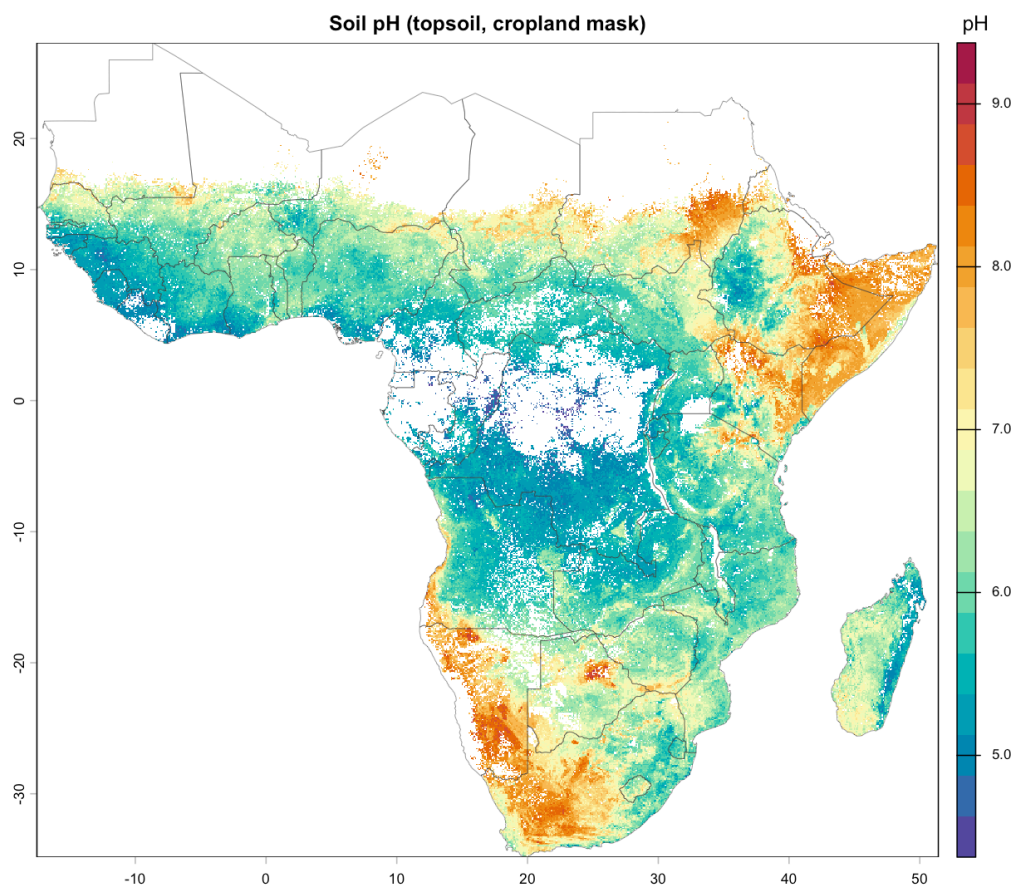


Figure 3: Topsoil pH across Sub-Saharan African cropland (SoilGrids-Africa, depth-weighted, ~9 km resolution).

6.2 Basalt application rate (B2 output)

Figure 4 shows the LITAS-year-1 basalt application rate, in tonnes per hectare. The map reproduces the pattern of the pH map closely, with two important modifications: rates are zero in alkaline soils (no liming need), and rates saturate at the agronomic maximum (~50 t/ha) in the most acidic pockets of the Congo basin and the Guinea coast.

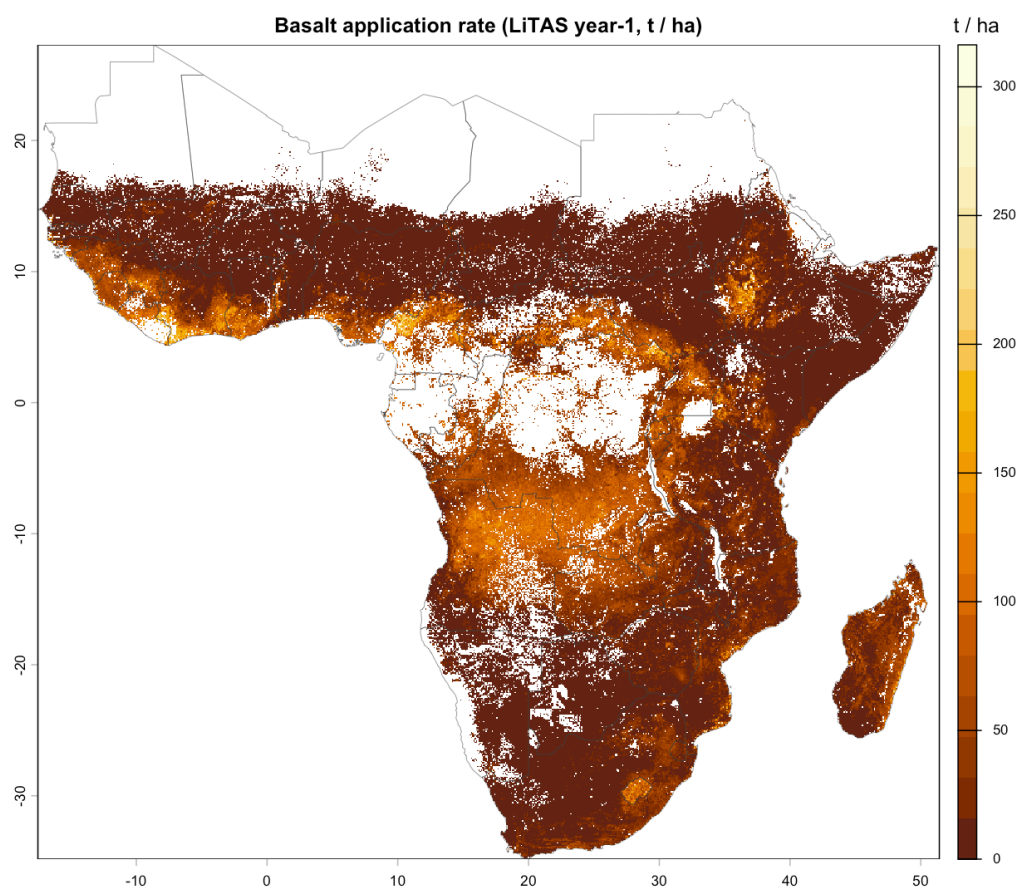


Figure 4: LiTAS year-1 basalt application rate (tonnes per hectare).

6.3 6.3 CDR yield (B3 output)

Figure 5 shows the per-hectare cumulative CDR yield under the LiTAS year-1 allocation. The map reflects the multiplicative composition of equations (3) and (4): high rates and high climate factors in the humid tropics produce the highest CDR yields (4–6 t CO₂ per hectare). The Sahel and the Horn of Africa are penalised by both the aridity-driven pedogenic-carbonate deduction and the low MAP-based climate factor; CDR there is below 1 t CO₂ per hectare even where small basalt-rate residuals exist.

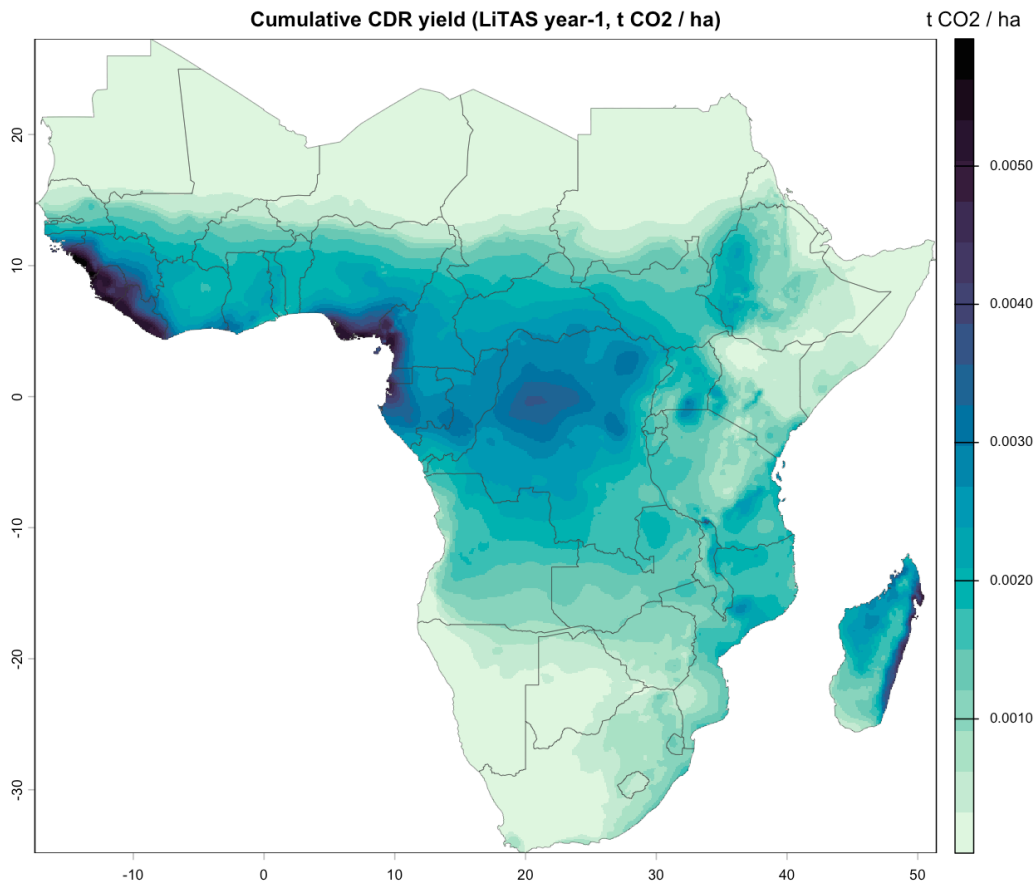


Figure 5: Cumulative CDR yield (t CO₂ per hectare, LiTAS year-1 allocation, climate × pH × pedogenic-C net).

6.4 6.4 Delivered basalt price (B4 output)

Figure 6 shows the delivered basalt price in US dollars per tonne. The pattern is dominated by the geography of basalt outcrops: prices are at the quarry-gate baseline (\$10 per tonne) in the Ethiopian highlands, the East African Rift, the Cameroon volcanic line, Madagascar, and southern Africa, and rise to the \$50 per tonne cap across West Africa and the Sahel. The cap binds for roughly half of pixels. This pattern is decisive for the economics: pixels near outcrops are agronomically attractive, biogeochemically efficient, *and* economically cheap, while pixels far from outcrops fail on the third dimension regardless of how favourable the first two are.

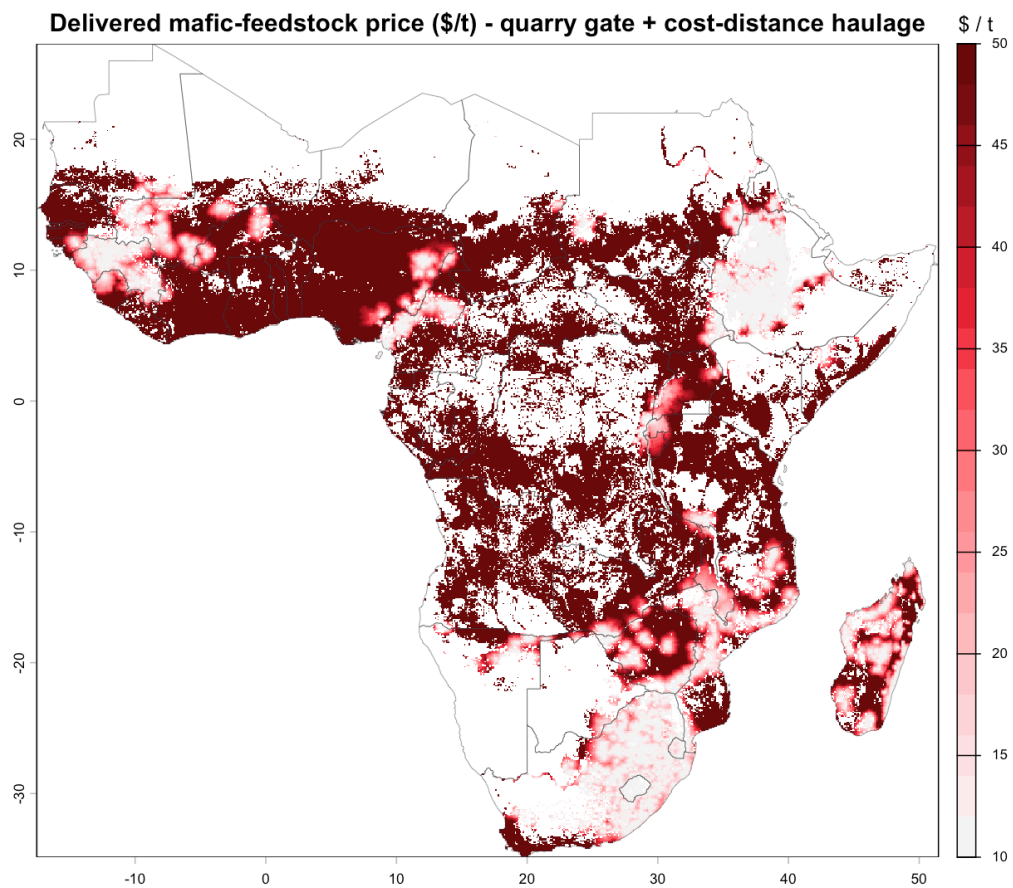


Figure 6: Delivered basalt price (\$ per tonne) — quarry-gate price plus cost-distance haulage over the MAP 2019 motorised friction surface.

6.5 6.5 Interpretation

Three patterns dominate the first-cut outputs.

First, the model's three input streams are spatially correlated but not identical. The intersection of high pH-driven lime demand, high climatic CDR efficiency, and low logistics cost is a relatively narrow band: roughly, Ethiopia, the East-African Rift, Madagascar, parts of southern Africa, and the Cameroon line. These regions are the natural early candidates for ERW deployment under our framework.

Second, the **delivered-price surface is the principal economic bottleneck**. The standard literature assumption of uniform \$30–50 per tonne delivered [10, 11] is reasonable as an SSA-wide average, but it masks a five-fold variation in real terms. Models that treat delivered price as a constant will both over-estimate deployment in West Africa and under-estimate it in East Africa.

Third, the **pedogenic-carbonate deduction is large** across a substantial fraction of the cropping area. In arid regions, the model deducts up to 90 % of the stoichiometric CDR potential. This is a deliberately conservative position but it is also the part of the model with the highest sensitivity to the choice of aridity proxy (Section 7.3).

7 7. Limitations and roadmap

We are explicit about the gaps in the current version. They fall into three categories.

7.1 7.1 Feedstock coverage

We currently consider only **basalt-equivalent mafic rocks** from the GLiM v1 classes vb (basic volcanic), py (pyroclastic, absent in Sub-Saharan Africa at GLiM resolution), and pb (basic plutonic, i.e. gabbro). The literature recognises four additional candidate-feedstock classes: **ultramafic silicates** (olivine, dunite, peridotite, serpentinite), **calcium-silicate by-products** (wollastonite, steel slag, cement-kiln dust), **construction by-products** (recycled concrete fines, fly ash), and **mine tailings** (especially nickel-mining and copper-mining tailings).

Ultramafic silicates are intentionally excluded for cropland use, following the recommendation of Beerling et al. [2] and Vienne et al. [10]: their nickel and chromium loads exceed agricultural-soil thresholds at typical application rates, and serpentinite carries asbestos risk. However, the **industrial alkaline by-products** — principally steel slag from blast-oxygen-furnace and electric-arc-furnace steelmaking, and cement-kiln dust from clinker production — are credible regional feedstocks that we currently omit. For some Sub-Saharan-African countries with no mafic outcrops in GLiM — in particular much of West Africa — a steel-slag-only or cement-kiln-dust-only feedstock supply may be the most economically realistic deployment pathway. Quantifying this requires a separate point-source inventory: locations of steel mills (e.g. from the Global Energy Monitor's Steel Plant Tracker), clinker plants (e.g. from the Global Cement Directory), and active mine tailings (national geological surveys), together with a per-source feedstock-chemistry table. We treat this as the highest-priority near-term extension.

7.2 7.2 Trial dataset

The Bayesian yield-uplift arm is currently fitted on 14 trial rows. This is a small dataset by any standard, and the posterior is prior-dominated. We have framed the model as a forward-compatible architecture that admits new trial evidence as it becomes available: the 2024 US Corn Belt trials [5], the 2024 Kisumu County trial [6], the 2025 InPlanet Brazil trials, and the 2025 Swiss vineyard trial [7] are all candidates for incorporation in the next iteration. The Kisumu trial in particular is structurally important because it is the only one that directly samples the agroecological zone (tropical smallholder, acidic soils, moderate-rainfall) in which most of our deployment-candidate pixels sit.

7.3 7.3 Pedogenic-carbonate proxy

We use mean annual precipitation alone as the aridity proxy in equation (4), binned into the UNEP-1992 categories. The pedogenic-carbonate literature generally uses the full aridity index $AI = MAP/PET$, where PET is potential evapotranspiration. Computing AI requires a PET surface, available from CGIAR-CSI or computable from WorldClim minimum-and-maximum temperatures via the Hargreaves equation. We expect this to be a small-effort, moderate-payoff upgrade and intend to make it in the next iteration.

7.4 7.4 Methodological caveats

Three further caveats are worth noting.

First, the **carbon-price assumption** of \$150 per tCO₂ is the 2024–25 mid-market for verified ERW credits. This is well above the regulated-market price for most CDR pathways and is contestable. The sensitivity sweeps cover \$0 through \$300; readers can re-rank pixels under their preferred price.

Second, the **MRV cost** is currently a flat \$30 per tCO₂. In practice MRV cost is scale-dependent and protocol-dependent; the Isometric and Puro protocols differ materially in their sampling cadence requirements. A future revision should differentiate MRV cost across countries and protocols.

Third, the model is **per-pixel and farmer-perspective**. A complete economic model of deployment would also incorporate **aggregator-level economics** — the costs of pooling MRV across many small farms, the costs of finance for distributed deployment, and the costs of coordinating with carbon-market intermediaries. These are non-trivial and likely to be material at the scale of smallholder-dominated cropping in Sub-Saharan Africa.

8 8. Conclusion

We have presented an ex-ante geospatial profitability model for basalt enhanced rock weathering across Sub-Saharan Africa. The model decomposes gross margin into three causal streams that converge on a single equation, exposes six decision variables for scenario analysis, and produces per-pixel indicator and outcome rasters at approximately one-kilometre resolution. First-cut illustrative outputs reproduce the qualitative geography expected from the literature, with three patterns dominating: a concentration of deployable land in the East-African and Cameroon volcanic zones, a five-fold spatial variation in delivered-feedstock price that the standard literature assumption of a uniform delivered price masks, and a large aridity-driven pedogenic-carbonate deduction in the Sahel.

The next iteration of the model will add an industrial-alkaline-by-product feedstock layer, an aridity-index pedogenic-carbonate calibration, an updated ERW-trial dataset, and a complete deployment run with sensitivity sweeps.

The model is intended as an open and continuously-updated piece of infrastructure rather than a finished product. Code, data, and documentation are available with this paper.

9 Acknowledgements

This work was completed under the *GAIA* project. We thank the field-trial investigators whose data underpins the Bayesian yield-uplift arm. Errors are our own.

10 References

All references below have been verified against the published record (May 2026). DOIs are given where available; trial preprints are linked to their canonical preprint server.

- [1] IPCC (2023). *Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Core Writing Team, H. Lee & J. Romero (eds.). IPCC, Geneva, Switzerland. Released 20 March 2023.
- [2] Beerling, D. J., Kantzas, E. P., Lomas, M. R., Wade, P., Eufrazio, R. M., Renforth, P., Sarkar, B., Andrews, M. G., James, R. H., Pearce, C. R., Mercure, J.-F., Pollitt, H., Holden, P. B., Edwards, N. R., Khanna, M., Koh, L., Quegan, S., Pidgeon, N. F., Janssens, I. A., Hansen, J., Banwart, S. A. (2020). Potential for large-scale CO₂ removal via enhanced rock weathering with croplands. *Nature* 583, 242–248. doi:10.1038/s41586-020-2448-9.
- [3] Strefler, J., Amann, T., Bauer, N., Kriegler, E., Hartmann, J. (2018). Potential and costs of carbon dioxide removal by enhanced weathering of rocks. *Environmental Research Letters* 13, 034010. doi:10.1088/1748-9326/aaa9c4.
- [4] Hartmann, J., West, A. J., Renforth, P., Köhler, P., De La Rocha, C. L., Wolf-Gladrow, D. A., Dürr, H. H., Scheffran, J. (2013). Enhanced chemical weathering as a geoengineering strategy to reduce atmospheric carbon dioxide, supply nutrients, and mitigate ocean acidification. *Reviews of Geophysics* 51, 113–149. doi:10.1002/rog.20004.
- [5] Beerling, D. J., Epihov, D. Z., Kantola, I. B., Masters, M. D., Reershemius, T., Planavsky, N. J., Reinhard, C. T., Jordan, J. S., Thorne, S. J., Weber, J., Val Martin, M., Freckleton, R. P., Hartley, S. E., James, R. H., Pearce, C. R., DeLucia, E. H., Banwart, S. A. (2024). Enhanced weathering in the US Corn Belt delivers carbon removal with agronomic benefits. *Proceedings of the National Academy of Sciences* 121 (9), e2319436121. doi:10.1073/pnas.2319436121.
- [6] Haque, F., Möller, B., Sagina, S., Odhiambo, C., Ondolo, H., Thuo, N., Kamau, K., Davies, S. (2025). *Agronomic Performance of Enhanced Rock Weathering in a Tropical Smallholder System: A Maize Trial in Kenya*. CDRXIV preprint 410 (Flux Carbon / UNCCD pilot), published 17 September 2025. <https://cdrxiv.org/preprint/410>.
- [7] Dupla, X., Bertagni, M. B., Grand, S. (2025). Three Years of Field Trials Indicate a Sustained Enhanced Rock Weathering Signal with Limited CO₂ Removal. *Environmental Science & Technology*

59 (48), 25751–25764. doi:10.1021/acs.est.5c09820. (Swiss vineyards; cited rate $\approx 100 \pm 30$ kg CO₂ ha⁻¹ yr⁻¹.)

[8] Kanzaki, Y., Planavsky, N. J., Zhang, S., Jordan, J. S., Suhrhoff, T. J., Reinhard, C. T. (2024). *Soil cation storage as a key control on the timescales of carbon dioxide removal through enhanced weathering*. ESS Open Archive preprint, doi:10.22541/essoar.170960101.14306457/v1. Accepted for *Environmental Research Letters* (May 2025).

[9] Renforth, P. (2019). The negative emission potential of alkaline materials. *Nature Communications* 10, 1401. doi:10.1038/s41467-019-09475-5.

[10] Vienne, A., Poblador, S., Portillo-Estrada, M., Hartmann, J., Ijehon, S., Wade, P., Vicca, S. (2022). Enhanced Weathering Using Basalt Rock Powder: Carbon Sequestration, Co-benefits and Risks in a Mesocosm Study With *Solanum tuberosum*. *Frontiers in Climate* 4, 869456. doi:10.3389/fclim.2022.869456.

[11] Lewis, A. L., Sarkar, B., Wade, P., Kemp, S. J., Hodson, M. E., Taylor, L. L., Yeong, K. L., Davies, K., Nelson, P. N., Bird, M. I., Kantola, I. B., Masters, M. D., DeLucia, E., Leake, J. R., Banwart, S. A., Beerling, D. J. (2021). Effects of mineralogy, chemistry and physical properties of basalts on carbon capture potential and plant-nutrient element release via enhanced weathering. *Applied Geochemistry* 132, 105023. doi:10.1016/j.apgeochem.2021.105023.

[12] Middleton, N. & Thomas, D. S. G. (eds.) (1992). *World Atlas of Desertification*. United Nations Environment Programme. Edward Arnold, London. 69 pp. (Used here for the MAP-keyed aridity bands.)

[13] Hengl, T., Mendes de Jesus, J., Heuvelink, G. B. M., Ruiperez Gonzalez, M., Kilibarda, M., Blagotić, A., Shangquan, W., Wright, M. N., Geng, X., Bauer-Marschallinger, B., Guevara, M. A., Vargas, R., MacMillan, R. A., Batjes, N. H., Leenaars, J. G. B., Ribeiro, E., Wheeler, I., Mantel, S., Kempen, B. (2017). SoilGrids250m: Global gridded soil information based on machine learning. *PLOS One* 12 (2), e0169748. doi:10.1371/journal.pone.0169748.

[14] Aramburu Merlos, F. (2022). *limer: Acidic soil management for agriculture*. R package, GAIA Africa. <https://github.com/gaiafrica/limer>.

[15] Aramburu Merlos, F., Silva, J. V., Baudron, F., Hijmans, R. J. (2023). Estimating lime requirements for tropical soils: Model comparison and development. *Geoderma* 432, 116421. doi:10.1016/j.geoderma.2023.116421. (Defines the LiTAS recipe used in erw-3.)

[16] Hijmans, R. J. & Manners, R. (2025). *Recocrop: Estimating Environmental Suitability for Plants*. R package version 0.4-2, CRAN. doi:10.32614/CRAN.package.Recocrop.

[17] You, L., Wood-Sichra, U., Fritz, S., Guo, Z., See, L., Koo, J. *MapSPAM — Spatial Production Allocation Model*. International Food Policy Research Institute (IFPRI) and HarvestChoice. Used here via `geodata::crop_spam`: SPAM 2010 V2.0 (global, 2019 release) for the global mask and SPAM 2017-SSA (2020 release) for Africa-specific data. <https://www.mapspam.info>.

[18] Bürkner, P.-C. (2017). brms: An R package for Bayesian multilevel models using Stan. *Journal of Statistical Software* 80 (1), 1–28. doi:10.18637/jss.v080.i01.

- [19] Weiss, D. J., Nelson, A., Vargas-Ruiz, C. A., Gligorić, K., Bavadekar, S., Gabrilovich, E., Bertozzi-Villa, A., Rozier, J., Gibson, H. S., Shekel, T., Kamath, C., Lieber, A., Schulman, K., Shao, Y., Qarkaxhija, V., Nandi, A. K., Keddie, S. H., Rumisha, S., Amratia, P., Arambepola, R., Chestnutt, E. G., Millar, J. J., Symons, T. L., Cameron, E., Battle, K. E., Bhatt, S., Gething, P. W. (2020). Global maps of travel time to healthcare facilities. *Nature Medicine* 26, 1835–1838. doi:10.1038/s41591-020-1059-1. (The motorised friction surface, nominal year 2019, used in erw-basalt-access is derived from this work — Malaria Atlas Project dataset_id Explorer__2020_motorized_friction_surface.)
- [20] Hartmann, J. & Moosdorf, N. (2012). The new global lithological map database GLiM: A representation of rock properties at the Earth surface. *Geochemistry, Geophysics, Geosystems* 13, Q12004. doi:10.1029/2012GC004370. Underlying dataset: *Global Lithological Map Database v1.0*, PANGAEA, doi:10.1594/PANGAEA.788537.